

Distribution of American Beech (*Fagus grandifolia*) and Beech Scale (*Cryptococcus fagisuga* Lind.) in Michigan from 2005 to 2009

Daniel J. Wieferich, Deborah G. McCullough, Daniel B. Hayes, and Nancy J. Schwalm

ABSTRACT

Beech scale (*Cryptococcus fagisuga* Lind.) (Hemiptera; Coccidae), an invasive insect associated with beech bark disease, was discovered in 2000 in a localized area of northwest Lower Michigan and one area in the eastern Upper Peninsula. From 2005 to 2009, we surveyed 809 sites in 73 Michigan counties to determine whether American beech (*Fagus grandifolia*) trees were present and, if so, whether trees had been colonized by beech scale. Beech trees were present in 696 sites in forest, recreation, and residential areas. By 2009, beech scale had infested beech trees in 8 of 15 counties in the Upper Peninsula, 14 of 68 counties in the Lower Peninsula, one of three islands in Lake Michigan, and all three sampled islands in Lake Huron. Colonization of islands more than 20 km from infestations on the mainland suggest that beech scale was likely introduced by humans or birds. Rates of expansion of 12 discrete beech scale populations varied from 3.3 to 14.3 km per year.

Keywords: beech bark disease, American beech, Michigan, invasive forest pest, *Fagus grandifolia*

Beech scale (*Cryptococcus fagisuga* Lind.) (Hemiptera; Coccidae) is an invasive insect associated with beech bark disease (BBD) on American beech (*Fagus grandifolia* Ehrhart) trees in eastern North America. Beech scale was introduced into Nova Scotia, Canada, around 1890 and has since expanded across much of the range of beech in the United States and Canada (Ehrlich 1934). Wounds created when scales pierce the bark of beech trees with their stylets to feed on sap enable two fungi (*Neonectria faginata* [Lohman, Watson, and Ayers] Castlebury and Rossman and *Neonectria ditissima* [Tulane and C. Tulane] Samuels and Rossman) to infect the trees (Ehrlich 1934, Castlebury et al. 2006). The *Neonectria* spp. fungi kill phloem and cambium tissue, causing lesions that can coalesce, disrupting water and nutrient transport within the tree (Ehrlich 1934, Houston 1994). Three phases of BBD have been defined (Shigo 1972, Houston and O'Brien 1983, Houston 1994). The "advancing front" refers to an area infested with beech scale but with little *Neonectria* spp. establishment. The second phase, known as the "killing front," refers to areas with abundant *Neonectria* infection and beech mortality. In the northeastern United States, approximately 50% of beech trees ≥ 25 cm in diameter typically die during the first wave of BBD (Houston 1994). The "aftermath forest" refers to areas where the BBD is well established and continually affects current and regenerating beech trees (Shigo 1972, Houston 1994).

Beech scales are parthenogenic, and populations can build rapidly (Ehrlich 1934). Scales reproduce annually in late summer and

fall, laying up to 50 eggs (Wainhouse and Gate 1988). After hatching, the mobile first instars, known as crawlers, move to a suitable location on their current tree or disperse to another host tree. Wind is thought to be the main vector of crawler dispersal (Felt 1933, Wainhouse 1980), although birds, wildlife, and human movement of infested logs and firewood may also introduce crawlers into new areas (Ehrlich 1934, Houston 1994, Morin et al. 2007). After settling, crawlers insert their stylets into the bark and begin feeding on the sieve cells of the secondary phloem (Frank Telewski, Department of Botany, Michigan State University, October 2009). Once feeding begins, the insects molt and become immobile. As scales feed, they secrete a white wax that makes heavily infested trees appear to be coated with white wool (Ehrlich 1934).

Trees infested by beech scale and infected by *N. faginata* and *N. ditissima* were first identified in 2000 in a campground in Mason County in the northwestern Lower Peninsula (L.P.) and in a campground in Luce County in the Upper Peninsula (U.P.) (McCullough et al. 2001, O'Brien et al. 2001). According to state and federal forest inventory data, approximately 15 million beech trees >22 cm in dbh were present in Michigan forests in 2000 (McCullough et al. 2001, Heyd 2005). These trees provide food and habitat for many species of wildlife and are economically valued in recreation areas as landscape trees and for lumber (Burns and Honkala 1990).

Delineating and monitoring the advancing front of BBD provides managers with the ability to incorporate potential impacts of

Manuscript received May 19, 2010, accepted March 10, 2011.

Daniel J. Wieferich (wieferi9@msu.edu), Department of Fisheries and Wildlife, Michigan State University, 13 Natural Resources, East Lansing, MI 48824. Deborah G. McCullough, Departments of Entomology and Forestry, Michigan State University, East Lansing, MI 48824. Daniel B. Hayes and Nancy J. Schwalm, Department of Fisheries and Wildlife, Michigan State University, East Lansing, MI 48824. We appreciate the cooperation provided by Robert Heyd, Michigan Department of Natural Resources. We thank James Wieferich, Jon Wagner, Justin Hodgins, Justin Miller, Pete Datema, Shayna Hart, and Will Folland (all from Michigan State University) and Nicholas Wieferich for assistance with fieldwork. This work was supported by funding from the US Forest Service Pest Trend-Impact Plot System (PTIPS) program.

This article uses metric units; the applicable conversion factors are: centimeters (cm): 1 cm = 0.39 in.; meters (m): 1 m = 3.3 ft; kilometers (km): 1 km = 0.6 mi; square kilometers (km²): 1 km² = 0.3861 mi².

Copyright © 2011 by the Society of American Foresters.

BBD into regeneration, harvest, or survey plans. Morin et al. (2007) mapped the distribution of beech using basal area data from the US Forest Service's Forest Inventory Analysis (FIA) database (Hansen et al. 1993) and then predicted the potential spread of BBD on the basis of its known distribution in 2003 and historical records. The FIA basal area data used to generate this map, however, were at a relatively coarse 1-km resolution (Morin et al. 2005) and excluded nonforested land types, such as urban, residential, and agricultural areas, where beech trees and BBD may occur. From 2001 to 2004, studies were conducted in Michigan to document beech scale distribution (Petrillo et al. 2005) and to collect baseline data on forest composition (Kearney et al. 2005). These two studies identified beech scale in 5 of 8 counties surveyed in the U.P. and 6 of 22 counties surveyed in the western and northern L.P.

We implemented an extensive survey across much of Michigan from 2005 to 2009 to identify sites with beech, determine which sites were infested with beech scale, and assess the progression of the advancing front. Distribution of beech scale was mapped, and the area colonized by beech scale was estimated annually.

Methods

From May to August of 2005, we surveyed 418 sites in 73 of Michigan's 83 counties. Sites were primarily on forested lands, but beech trees in agricultural and urban landscapes were also examined. We initially used the beech basal area map (1-km resolution) produced by Morin et al. (2005) and a spatial land use layer provided by the Michigan Department of Natural Resources (MDNR) to locate areas likely to contain beech trees. The MDNR spatial layer (30 m resolution) known as the Integrated Forest Monitoring Assessment and Prescription (IFMAP) delineated areas of northern hardwood cover. This cover type was deemed most likely to include beech by state forestry and forest health specialists (MDNR 2003). Areas likely to contain beech, based on the beech basal area map or IFMAP, were divided into 10-km² sections, and we sampled one beech stand per section. Additional areas, however, were also explored to ensure a high proportion of the beech resource was represented in our survey. In these areas, we attempted to sample one beech stand per 30 km². We focused our search on wooded areas likely to contain beech. When beech was not found within a 30 km² area, we recorded a "no beech" site. We did not survey sites in 10 counties that are highly urbanized, predominantly agricultural, or beyond the range of beech (e.g., the western Upper Peninsula). We calculated the proportion of our beech sites that corresponded with the IFMAP northern hardwood cover type by overlaying our sites with the IFMAP layer in ESRI ArcMap.

Stands were initially scanned for beech tree presence from the roadside. If the habitat appeared suitable but no beech trees were visible from the road, we walked through the stand. On public land or accessible private land, we established a plot center where we first encountered at least three beech trees ≥ 12 cm dbh. Branches and trunks of beech trees within a variable-radius plot delineated with a 10 basal area factor angle gauge (prism) were carefully examined to determine whether beech scale was present. A second crew member subsequently examined the trees to ensure that even light infestations of scale were detected. In several cases, a single infested tree with only a trace population of beech scale was identified, which suggests that our visual surveys effectively detected relatively light infestations. Beech trees were similarly examined in an additional plot established 100 m to the north or south of the center plot, and a second plot 100 m to the east or west. The directions for the second

and third plots were randomly selected, unless property boundaries limited access. In these cases, we rerandomized and selected from accessible directions. In areas with limited or minimal access and in nonforested areas (e.g., roadside parks, campgrounds, landscape trees) we examined the first 10 beech trees encountered to determine whether beech scale was present. When fewer than 10 beech trees were available, all available beech trees were sampled.

We delineated the advancing front of BBD each summer (May to August) using two techniques. First, we established additional sites in concentric radii, 1–5 km from sites determined to be infested with beech scale on the advancing front. Additionally, from 2006–2009, we revisited the previously established sites located in close proximity to the advancing front to determine whether the sites had become infested. When beech scale was discovered at revisited sites, the next closest previously uninfested site was visited. This process continued until a buffer of uninfested sites was established around infested sites. If few sites with beech trees existed along the advancing front, an adaptive sampling technique was used to fill gaps in the sampling distribution. This technique involved establishing new site locations within 10 km of sites known to be infested and in between the advancing front and the closest uninfested sites. Once beech scale infestation was observed, we assumed the site would remain infested, and those sites were not resurveyed.

Between 2005 and 2009, a total of 803 sites were examined, and 254 sites were revisited. The sites were distributed across 11 of the 15 U.P. counties and 62 of the 68 L.P. counties. In 2006, 48 sites that were uninfested in 2005 were revisited, and 231 new sites were established. In 2007, 43 sites that were uninfested in 2006 were revisited, and 52 new sites were established. In 2008, 60 sites that were uninfested in 2007 were revisited, and 21 new sites were established. In 2009, 103 sites that were uninfested in 2008 were revisited, and 81 new sites were established.

Beech trees on six islands were examined at least once between 2006 and 2009, including Beaver, North Manitou, and South Manitou Islands in Lake Michigan and Bois Blanc, Drummond, and Mackinac Islands in Lake Huron. These islands range from roughly 14.5 to 645 km² in size and are < 2 to 24 km from mainland (Table 1). Sites on islands were sampled using the same methods as on the mainland. Time and funding constraints did not permit us to visit islands every year.

Results from our sampling were imported into ESRI ArcGIS 9.2. We defined disjunct populations as beech scale infestations separated geographically from one another by at least 20 km of uninfested area containing some beech component. We selected the 20-km threshold because this distance is farther than the natural spread of beech scale, estimated to be 14.7 ± 0.9 km per year (Morin et al. 2007). Each disjunct population was assigned a name corresponding to the county where it originated or was most abundant (although some populations spanned multiple counties). Using the Hawth's Analysis Tools extension in ArcMAP (Beyer 2004), the total area of beech scale infestation was calculated annually for each disjunct population using the minimum convex polygon (MCP) method. Mohr (1947) defines an MCP as the complete enclosure of all data points by connecting the outer locations to create a convex polygon. Areas of minimum convex polygons for consecutive years were compared to evaluate annual change in beech scale distribution. We also estimated maximum annual spread rates by measuring the maximum spread between the advancing front of each population between 2006 and 2009, then dividing the distance by the 3-year time frame.

Table 1. Characteristics of Michigan islands sampled for beech scale (*Cryptococcus fagisuga* Lind.) presence between 2005 and 2009. The year of sampling on each island is shown, along with total number of plots, number of infested plots, and area of infestation (km²) by year.

Island	No. of plots	No. of infested plots	Area infested (km ²)	Lake	Residents ^a	Distance (km) ^b	Area (km ²) ^c
Beaver				Michigan	Yes	24	144.5
2005	12	3	16				
2006	15	5	36				
Bois Blanc				Huron	Yes	6	127
2006	9	4	4				
Drummond				Huron	Yes	<2	645
2006	10	0	0				
2008	9	2	NA ^d				
2009	8	3	12				
Mackinac				Huron	Yes	4	14.5
2006	13	7	3				
N. Manitou				Michigan	No	11	22
2005	14	0	0				
S. Manitou				Michigan	No	11	21
2005	3	0	0				
2006	6	0	0				

^a Indicates whether residents live on the island.

^b Distance represents the shortest straight-line distance to a beech scale infestation on Michigan's mainland.

^c Island size.

^d NA, Infested points were too scarce to create the minimum convex polygon.

Results

Beech Tree Distribution

Beech trees were present in 696 of the 803 (86.5%) sites surveyed between 2005 and 2009 (Figure 1). Beech occurred in 49 counties in the L.P. and 8 counties in the eastern and central U.P. Beech trees were present on all six of the sampled islands.

There was little overlap between the sites with beech trees and the northern hardwoods layer of IFMAP. Only 201 of the 696 (29%) sites with beech trees occurred in areas classified as northern hardwoods. The remaining 495 sites with beech were generally located in areas designated by IFMAP as roads (20.5%), aspen association (19%), agriculture (16%), oak association (12%), or upland mixed forests (6%). Extensive areas designated as northern hardwoods forest type occur in the western U.P., but the range of beech does not extend into this area.

We also qualitatively compared beech presence and absence in our sites with the beech map generated with FIA data by Morin et al. (2005). More than 90% of the 696 sites where we found beech corresponded with their map of beech presence. The additional sites where we observed beech were nearly all in agricultural or urban settings, where FIA data are not collected (Burkman 2005).

Beech Scale Distribution in the Lower Peninsula

In 2005, beech scale was distributed across substantial areas in the northern and western portions of Michigan's L.P. (Figure 2 and Table 2). By 2009, a total of 145 of the 492 sites (29%) with beech trees were infested, encompassing a total of 2667 km² in three distinct populations located in Wexford, Oceana, and Emmet Counties (Figures 1 and 2). Beech trees in the eastern and southern L.P. remained uninfested as of 2009.

Beech scale distribution increased annually from 2005 to 2009 in the L.P., particularly between 2006 and 2007, when the total infested area increased by 35% (Table 2). Newly established, disjunct populations were detected in an isolated group of beech trees in a forested site in Charlevoix County in 2006, and two more distinct populations, in Crawford and Cheboygan Counties, were discovered in 2009 (Figure 2). Only one infested site was detected in each of the Charlevoix and Cheboygan populations. After discovering the

Cheboygan scale population in 2006, areas with beech trees within 1 km² were closely monitored annually, along with our regular sampling, but we found no evidence of additional infestations by 2009. We also intensified sampling within 1 km² of the Cheboygan population in 2009 but, similarly, did not detect additional infestations. Based on these efforts, we assume that both Charlevoix and Cheboygan populations remained <1 km² in 2009 (Table 2). Overall, the infested area in the L.P. expanded by 146% during the 5-year period. The rate of expansion, however, varied among individual populations (Table 2). For example, between 2005 and 2008, the Oceana infestation expanded by 40%, with a maximum spread rate of 3.3 km per year, whereas the Wexford infestation increased in area by 290%, with a maximum spread rate of 8 km per year. These two infestations coalesced in 2009. In Emmet County, the infestation increased from 77 km² in 2005 to 841 km² in 2008 and 1,155 km² in 2009, yielding a maximum spread rate of 14.3 km per year. In contrast, the Charlevoix infestation did not expand in area between 2006 and 2009.

Beech Scale Distribution in the Upper Peninsula

In 2005, beech scale was present in an area encompassing approximately 6214 km² in the U.P. (Figure 2). Unlike the discrete infestations in the L.P., the beech scale population in the U.P. was contiguous until 2009, when a localized infestation was found in Menominee County, approximately 45 km southwest of the main population (Figure 2). The main infestation increased by 10 to 24% annually between 2006 and 2008, with a maximum spread rate of 11 km per year (Table 2). By 2009, beech scale was present in 84 of the 139 sites (60%) with beech, which included nearly all forests with a beech component in the eastern and central U.P. Including the Menominee population, the total infested area encompassed 11,373 km² by 2009 (Figure 1 and Table 2).

Beech Scale Distribution on Islands

In 2005, approximately 16 km² of Beaver Island was infested with beech scale (Table 1). In 2006, 10 of the 12 sites from 2005 were revisited, and 4 new sites were created to fill gaps in our sampling. All three previously infested sites remained infested, three of

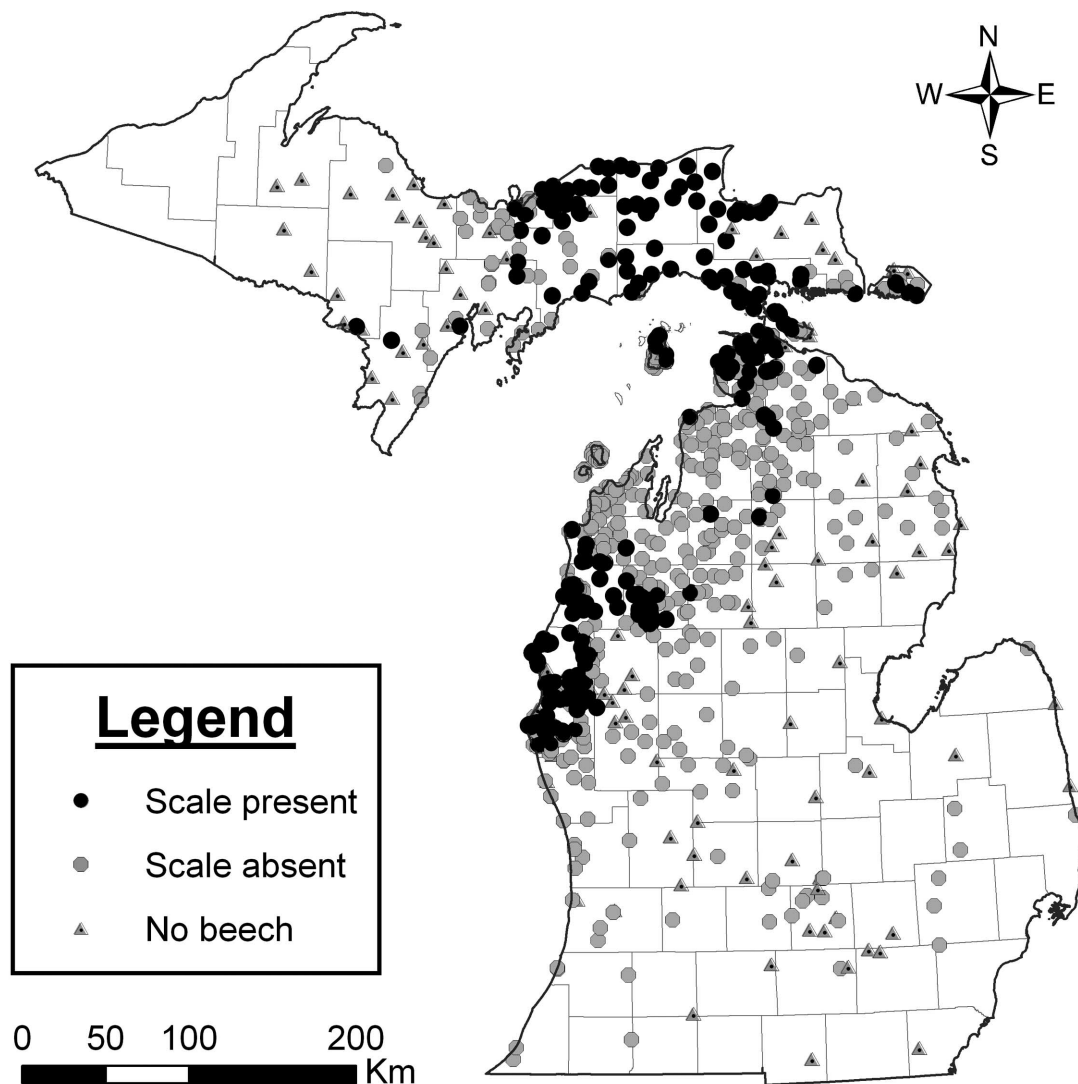


Figure 1. Locations of beech trees and beech scale (*Cryptococcus fagisuga* Lind.) in surveys conducted from 2005 to 2009.

the eight previously uninfested sites became infested, one new site contained beech but no scale, and three new sites had no beech trees. The new infestations occurred in the sites closest to previous infestations. Total infested area increased 125% from 2005 to 2006 (Table 1).

Eight sites with beech and two without beech were established on Drummond Island in 2006, but beech scale was not detected (Table 1). In 2008, beech trees in these eight sites and one additional site were examined. Beech scale was present in two of the nine sites, both located in the south-central portion of the island. A sample of two infested sites was inadequate to estimate the infested area using the minimum convex polygon method. In 2009, beech scale was detected at two previously uninfested sites, located in close proximity to sites infested in 2008 (Table 1).

Beech scale was present in 2006 on both Bois Blanc and Mackinac Islands. Of the nine sites examined on Bois Blanc, four sites on the west side of the island had beech scale (Table 1). Seven of 13 sites established on Mackinac Island had beech scale. Unlike the other islands, infested sites were scattered throughout the island, encompassing 3 km².

In 2005, 15 sites on North Manitou Island and 3 sites on South Manitou Island were sampled, but there was no evidence of beech

scale on either island (Table 1). Six additional sites were surveyed on South Manitou Island in 2006, but beech scale was not detected on the island in 2006.

Discussion

We originally expected the relatively high resolution IFMAP data could be used to efficiently identify areas likely to contain beech trees. Beech trees occurred in several cover types other than northern hardwoods, however, and much of the area mapped as northern hardwoods had no beech component. In contrast, presence and absence of beech recorded during our surveys consistently overlapped with a map of beech basal area generated by Morin et al. (2005) using FIA data, despite the relatively coarse scale of the FIA data (1 km). Both sets of data were of limited value in areas dominated by agriculture or urban land types, although beech trees were often present in these sites.

Previous efforts to document beech scale distribution and assess impacts of BBD were initiated relatively soon after it was discovered in Michigan in 2000. Petrillo et al. (2005) surveyed 264 sites from 2001 to 2003 in 8 counties in the U.P., 22 counties in the western and northern L.P., and 5 islands. They detected beech scale in five eastern U.P. counties and four counties in the west-central region of

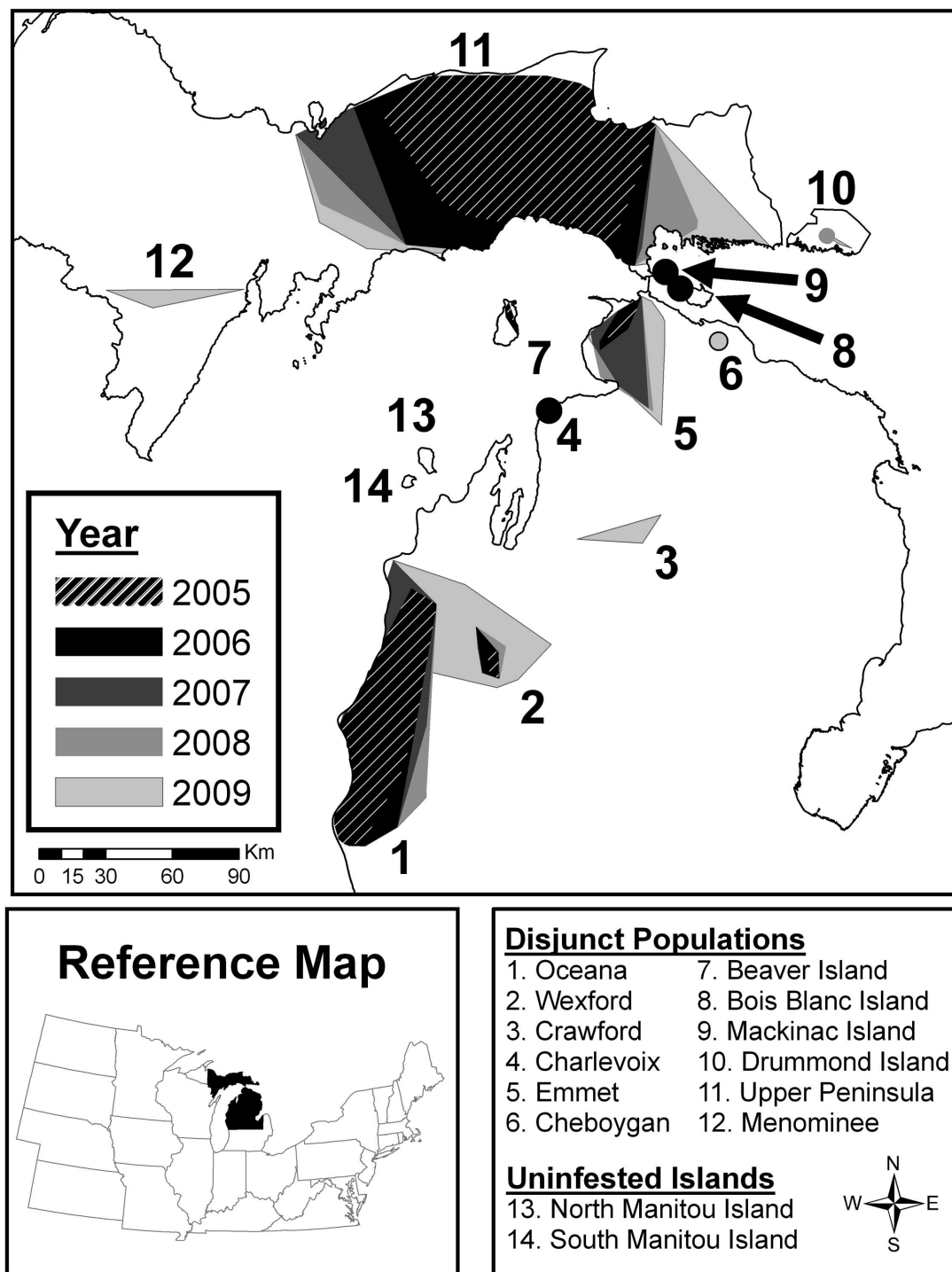


Figure 2. Areas with beech scale (*Cryptococcus fagisuga* Lind.) populations as delineated by minimum convex polygons annually from 2005 to 2009.

the L.P. No evidence of scale was observed on the islands. Our 5-year study, which encompassed 803 total sites and 254 revisits across 73 counties and 6 islands, allowed us to assess expansion of beech scale over time and at a relatively fine scale. We detected nine disjunct populations not previously documented, including Charlevoix, Cheboygan, Crawford, Emmet, and Menominee Counties, along with Beaver, Mackinac, Bois Blanc, and Drummond Islands. In addition, our data showed that populations identified in previous studies had increased substantially in area.

The distribution of beech scale we mapped in 2009 was similar to that predicted by Morin et al. (2007), who modeled beech scale spread from 2003 to 2010 using historical data. They noted that from 1911 to 2003, the advancing front of BBD in North America spread 14.7 ± 0.9 km per year. Using data from sites surveyed between 2005 and 2007, Schwalm (2009) determined beech scale advanced by 4 and 1.5 km per year in the U.P. and L.P., respectively. Our data, however, suggest maximum spread rates of individual populations were as high as 11 and 14.3 km per year in the U.P. and L.P., respectively.

Table 2. Area (km²) of disjunct populations of beech scale (*Cryptococcus fagisuga* Lind.) infestations, as determined by the minimum convex polygon method, by year. Dashes indicate that the population was not sampled in the corresponding year.

Disjunct population	Year				
	2005	2006	2007	2008	2009
Total Lower Peninsula	2,667	3,098	4,170	4,600	6,551
Oceana	2,539	2,813	3,287	3,560	5,200 ^a
Wexford	51	136	144	199	
Emmet	77	149	739	841	1,155
Crawford	0	—	—	—	196
Charlevoix ^b	0	<1	<1	<1	<1
Cheboygan ^b	0	—	—	—	<1
Total Upper Peninsula	6,214	8,203	9,187	10,373	11,802
Upper Peninsula	6,214	8,203	9,187	10,373	11,547
Menominee	0	—	—	—	255

^a The Wexford and Oceana populations coalesced in 2009.

^b Disjunct populations defined by a single infested site; area could not be estimated with the minimum convex polygon method. Intensive sampling indicated that these infestations were extremely localized and <1 km² in area.

Although we used MCPs to calculate distributions of disjunct populations using our presence data, other techniques, including inverse distance weighting and kriging, that use presence and absence data, were also explored. Because our sampling distribution varied among years, however, annual comparisons of these estimations would be inappropriate. Although our observations indicate the MCPs provided reasonable estimates of beech scale range, one shortcoming of this method is that distribution can be overestimated, especially for irregular shapes (Burgman and Fox 2003). This occurred in some disjunct populations where sites with unin-fested beech trees fell within a polygon, as defined by the MCP analysis. Ideally, in such situations, additional sites in close proximity to the uninfested areas should be sampled to better define the polygon and ensure the advancing front is accurately delineated. If enough sites are present, subsets of sites within a population can be used to create multiple MCPs per population to better delineate areas of concave distributions.

Spread rates varied among disjunct populations and years, perhaps reflecting variation in the continuity of beech distribution, as well as fluctuations in weather conditions. The Oceana population, surrounded by urban and agricultural lands with a fragmented distribution of beech, spread relatively slowly. In contrast, infestations in Wexford and Emmet Counties occurred in the midst of a relatively continuous distribution of beech, and scale populations expanded relatively rapidly.

The numerous disjunct populations in the L.P. suggest that multiple mechanisms of scale dispersal may occur. Previous studies reported that wind dispersal of crawlers is the primary mode of natural dispersal (Felt 1933, Wainhouse 1980). At wind speeds of less than 2 m/second, crawlers dispersed an average of 10 m, limited by the elevation of the insect and wind speed. Wainhouse and Gate (1988) suggested that scale crawlers above the canopy could be transported for longer distances in high winds. Other agents, including migratory birds, wildlife, and human transport of infested logs or firewood could also contribute to beech scale dispersal (Ehrlich 1934, Houston 1994, Morin et al. 2007). We identified 12 discrete beech scale infestations located >20 km apart, suggesting that wind dispersal was not responsible. We also observed a new infestation of beech scale become established in 2006 at a site immediately adjacent to a sawmill that harbored infested beech logs during the summer and early fall. This infestation was approximately 10 km from the nearest infested beech tree. Although long-distance dispersal of beech crawlers by wind is possible, it seems more likely that the infested logs were the source of the crawlers that initiated the infestation.

Monitoring beech scale on six islands also provided insights into mechanisms of spread, especially anthropogenic actions. Although the islands differ in size and distance from infestations on the mainland, by 2009, only the Manitou Islands remained uninfested. Unlike other Michigan islands, neither of the Manitou Islands have permanent residents and visitors, and firewood is closely monitored by park personnel. Beaver Island is 24 km from the nearest beech scale infestation on the mainland, and wind dispersal of scale crawlers appears unlikely. Although bans on transport of firewood were recently implemented, residents or tourists could have previously introduced beech scale onto islands. Also, several residents have private transportation, making enforcement of the ban difficult. Beaver Island is also located in flyways of several migratory bird species (Lincoln 1939). Crawlers could conceivably have been transported to Beaver Island in late summer or early fall from infested sites in Upper Michigan.

Monitoring beech scale distribution over time can allow natural resource managers to incorporate BBD into management plans. Current beech scale distribution data can be used to guide timber sales, firewood harvest, or log transport to reduce the risk of transporting beech scale to uninfested areas when crawlers are present. Reducing overstory beech density and overmature beech in areas adjacent to infestations may help minimize the spread of beech scale and BBD impacts (Mielke et al. 1987).

The spread of beech scale in Michigan, reflecting a combination of local and long-distance dispersal, appears similar to previous observations in the northeastern region of the United States. Infestations of beech scale were discovered in Massachusetts, Maine, New Hampshire, New York, and New Jersey in 1935; in West Virginia in 1990; in North Carolina and Tennessee in 1999; in Ohio and Michigan in 2000; and in northeastern Wisconsin in 2009 (Houston 1994, O'Brien et al. 2001, Wisconsin Department of Natural Resources Forestry Division 2009). Each of these populations met our criterion of being disjunct populations. Our data confirm that long-distance dispersal is occurring throughout Michigan too, as evidenced by the development of several distinct disjunct populations between 2000 and 2009. Given the ability of beech scale to spread at variable rates, the remaining states with uninfested beech trees should consider monitoring their resource and preparations should be made for beech scale infestations in the near future.

Literature Cited

BEYER, H.L. 2004. *Hawth's analysis tools for ArcGIS*. Available online at www.spatialecology.com/htools; last accessed Oct. 15, 2010.

- BURGMAN, M.A., AND J. FOX. 2003. Bias in species range estimates from minimum convex polygons: Implications for conservation and options for improved planning. *Animal Conserv.* 6:19–28.
- BURKMAN, B. 2005. *Forest Inventory and Analysis: Sampling and plot design*. US For. Serv. Available online at www.fia.fs.fed.us/library/fact-sheets/datacollections/Sampling%20and%20Plot%20Design.pdf; last accessed May 25, 2011.
- BURNS, R.M., AND B.H. HONKALA. 1990. *Silvics of North America: 1. Conifers; 2. Hardwoods*. US For. Serv. Agriculture Handbook 654(2). 877 p.
- CASTLEBURY, L.A., A.Y. ROSSMAN, AND A.S. HYTEN. 2006. Phylogenetic relationships of *Neonectria/Cylindrocarpon* on *Fagus* in North America. *Can. J. Bot.* 84:1417–1433.
- EHRLICH, J. 1934. The beech bark disease: A *Nectria* disease of *Fagus*, following *Cryptococcus Fagi* (Baer.). *Can. J. Res.* 10:593–690.
- FELT, E.P. 1933. The beech scale and a fungus. *J. Econ. Entomol.* 26:510.
- HANSEN, M.H., T. FRIESWYK, J.F. GLOVER, AND J.E. KELLY. 1993. *The eastwide forest inventory data base: Users Manual*. US For. Serv. Gen. Tech. Rep. NC-151. 22 p.
- HEYD, R.L. 2005. Managing beech bark disease in Michigan. P. 128–132 in *Beech bark disease: Proceedings of the beech bark disease symposium*, Evans, C.A., J.A. Lucas, and M.J. Twery (eds.). US For. Serv. Gen. Tech. Rep. NE-331. US For. Serv., Northeast. Res. Stn., Newtown Square, PA.
- HOUSTON, D.R. 1994. Major new tree disease epidemics: Beech bark disease. *Annu. Rev. Phytopathol.* 32:75–87.
- HOUSTON, D.R., AND J.T. O'BRIEN. 1983. *Beech bark disease*. US For. Serv. Forest Insect and Disease Leaflet 75. Available online at www.fs.fed.us/r6/nr/fid/fidls/fidl-75.pdf; last accessed Oct. 17, 2010.
- KEARNEY, A., D.G. MCCULLOUGH, AND M. WALTERS. 2005. Impacts of beech bark disease on wildlife resource abundance in Michigan. P. 92–93 in *Beech Bark Disease: Proceedings of the Beech Bark Disease Symposium*, Evans, C.A., J.A. Lucas, and M.J. Twery (eds.). US For. Serv. Gen. Tech. Rep. NE-331. US For. Serv., Northeast. Res. Stn., Newtown Square, PA.
- LINCOLN, F.C. 1939. *The migration of American birds*. Doubleday Doran and Company, New York. 189 p.
- MCCULLOUGH, D., R.L. HEYD, AND J.G. O'BRIEN. 2001. *Biology and management of beech bark disease*. Michigan State University Extension Bulletin E-2746.
- MICHIGAN DEPARTMENT OF NATURAL RESOURCES, FOREST, MINERAL, AND FIRE MANAGEMENT DIVISION. 2003. *IFMAP/GAP Lower Peninsula land cover*. Available online at www.dnr.state.mi.us/spatialdatalibrary/sdl2/land_use_cover/2001/IFMAP_lp_landcover.htm; last accessed Oct. 17, 2010.
- MIELKE, M.E., D.R. HOUSTON, AND A.T. BULLARD. 1987. Beech bark disease management alternatives. P. 272–280 in *Proc. of Integrated pest management symposium for northern forests*, Cooperative Extension Service, University of Wisconsin–Extension, Madison, WI.
- MOHR, C. 1947. Table of equivalent populations of North American mammals. *Am. Midland Nat.* 37:223–249.
- MORIN, R., A. LIEBHOLD, E. LUZADER, A. LISTER, K. GOTTSCHALK, AND D. TWARDUS. 2005. *Mapping host-species abundance of three major exotic forest pests*. US For. Serv. Res. Pap. NE-726. 11 p.
- MORIN, R., A. LIEBHOLD, P. TOBIN, K. GOTTSCHALK, AND E. LUZADER. 2007. Spread of beech bark disease in the eastern United States and its relationship to regional forest composition. *Can. J. For. Res.* 37:726–736.
- O'BRIEN, J.G., M.E. OSTRY, AND M.E. MIELKE. 2001. First report of beech bark disease in Michigan. *Plant Disease* 69:905.
- PETRILLO, H., J.A. WITTER, AND E.M. THOMPSON. 2005. Michigan beech bark disease monitoring and impact analysis system. P. 48–51 in *Beech bark disease: Proceedings of the beech bark disease symposium*, Evans, C.A., J.A. Lucas, and M.J. Twery (eds.). US For. Serv. Gen. Tech. Rep. NE-331. US For. Serv., Northeast. Res. Stn., Newtown Square, PA.
- SCHWALM, N. 2009. *The beech scale (Cryptococcus fagisuga) in Michigan: distribution, models of spread and relation to forest and wildlife resources*. MSc thesis, Michigan State Univ., East Lansing, MI. 147 p.
- SHIGO, A.L. 1972. The beech bark disease today in the northeastern United States. *J. For.* 70:286–289.
- WAINHOUSE, D. 1980. Dispersal of first instar larvae of the felted beech scale, *Cryptococcus fagisuga*. *J. Appl. Ecol.* 17(3):523–532.
- WAINHOUSE, D., AND I.M. GATE. 1988. The beech scale. P. 76–85 in *Dynamics of forest insect populations*, A.A. Berryman (ed.). Plenum, New York.
- WISCONSIN DEPARTMENT OF NATURAL RESOURCES FORESTRY DIVISION. 2009. *Other exotic forest threats: Beech bark disease*. Available online at dnr.wi.gov/forestry/fh/exotics/exotic-bb.htm; last accessed May 25, 2011.